2024 5th IEEE Global Conference for Advancement in Technology (GCAT)
Karnataka, India. Oct 4-6, 2024

Enhancing Kidney Disease Diagnosis through Advanced Deep Learning Techniques: A Comprehensive Analysis in Modern Medical Imaging

Pratham Kaushik
Centre for Research Impact & Outcome, Chitkara University
Institute of Engineering and Technology
Chitkara University,
Rajpura-140401, Punjab, India
pratham.kaushik@chitkara.edu.in

Pooja Sharma
Chitkara Centre for Research and Development,
Chitkara University,
Himachal Pradesh-174103 India
pooja.sharma.orp@chitkara.edu.in

2024 5th IEEE Global Conference for Advancement in Technology (GCAT) | 979-8-3503-7668-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/GCAT62922.2024.10924107

***Abstract*— This research work develops an effective approach for diagnosing kidney diseases by applying CNNs. The research focuses on classifying four specific kidney conditions—normal, cyst, tumor, and stone—by training and testing three CNN models: This includes EfficientNetB3, MobileNetV2, and ResNet50. The models were trained on a large set of CT images, and it was noted that the performance parameters were quite high. Among all the EfficientNetB3 models, provided the highest overall accuracy of 90.46%, this indicates greater efficiency in precision and recall in all the categories presented. MobileNetV2 and ResNet50 achieved considerable scores, specifically 89% and 89.50% of accurate predictions, respectively, although there were small discrepancies in their measurements, especially in distinguishing cysts and stones. Resultantly, the study highlights the possibility of CNNs to enhance diagnostic accuracy of kidney disease and consequently support early identification, clinical management, and improved patient safety and prognoses.**

Keywords: Kidney Disease Detection, Medical Imaging, Convolutional Neural Networks (CNNs), Machine Learning in Healthcare, Kidney Condition Classification, Deep Learning Models.

I. INTRODUCTION

Chronic kidney disease, also known as CKD, is a global health problem that affects millions of patients and is characterized by a gradual and asymptomatic decline in kidney function. The identification and treatment of CKD at an early stage is critical in minimizing the serious consequences, such as ESRD; however, this is easier said than done. CKD is difficult to diagnose because in most cases the disease is asymptomatic during the early stages, and thus doctors are unable to detect the disease in its early stages when its effects are not severe [1]. Conventional markers include serum creatinine and the estimated glomerular filtration rate (eGFR), which are effective but not very specific for early CKD identification [2]. Such a delay can result in the best chance to intervene early, and this is where the new technologies come in handy. Over the past few years, the application of machine learning and artificial intelligence has gained much concern in diagnosing and managing CKD. These new advanced technologies can help to analyze the huge amount of patients' data and predict further development of the diseases, which is impossible with the usual methods [3]. In the same way, machine learning models could be trained to detect the small changes in the kidney function that may signal the beginning of CKD and therefore be diagnosed at an early stage [4]. This is more crucial in managing the CKD because its early intervention helps to reduce the rate of progression and minimize complications [5]. The subject of this research is to study how some selected CNNs, including EfficientNetB3, MobileNetV2, and ResNet50, can be utilized to enhance CKD diagnosis. These models have proved to be efficient in other medical image analysis tasks; therefore, their application in CKD can result in accurate and early diagnosis. These models will be analysed with the purpose of determining which of these methods could prove most effective in the early detection of CKD, thus improving the outcomes of patients and reducing, where possible, the suffering from this chronic illness.

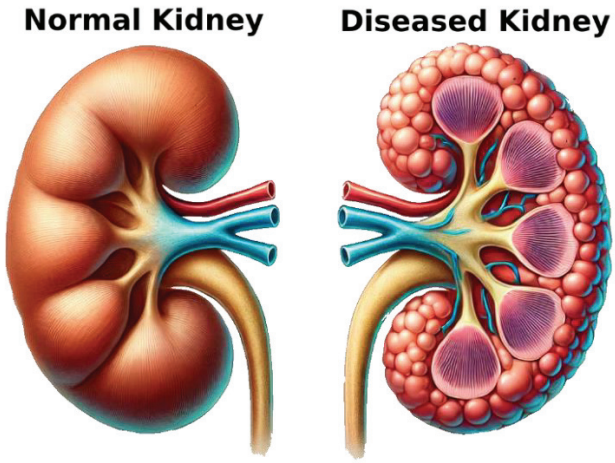

Fig. 1. Normal Kidney vs Diseased Kidney

II. LITERATURE REVIEW

The diagnosis of CKD has been made using deep learning and machine learning techniques in the recent past. Other aspects related to the CKD diagnosis are described by Kalantar-Zadeh et al. [6], where the authors present the global epidemiology of CKD and the demand for the enhanced diagnostic methods. These changes summarized in the present studies reveal the shift in the diagnosis of CKD, which is marked by the incorporation of the latest technologies like machine learning, deep learning, and genetic tests. Qezelbash-Chamak et al. [7] have also further extended the use of machine learning in CKD diagnosis by presenting a comprehensive review about the complete use of different machine learning methods in diagnosing kidney diseases. For example, Sugahara et al. [8] described the challenges in diagnosing DKD, which is one of the leading CKD types, and described how machine learning models can be used to differentiate DKD from other forms of kidney diseases. In the same respect, Murugesan et al. [9] showed that the fuzzy logic-based systems are effective in handling the uncertainty

Authorized licensed use limited to: MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on November 28,2025 at 10:52:17 UTC from IEEE Xplore. Restrictions apply.

associated with medical data and therefore improve the accuracy of CKD diagnosis in clinical practice. However, one of the most significant developments in CKD management is genetic testing because of hereditary diseases. In their study, Knoers et al. [10] underlined that the genetic testing is useful for early diagnosis of specific forms of CKD that require an individualized therapeutic approach. This is supported by the work of Rahman et al. [12], who provide insight on how predictive modelling using machine learning can help predict the progression of CKD by considering genetic and clinical factors in the management of CKD. Other deep learning models, such as the CNNs, have also played a significant role in the development of CKD diagnosis. Jain et al. [14] and Preethi et al. [15] explored the adoption of CNN-SVM models and optimized deep learning algorithms, respectively, with adequate applications on CKD identification and prediction. These studies will, therefore, prove that deep learning can change the face of diagnosis in CKD by making it far more accurate and accessible. This could also be extended to a few new ways that researchers now develop and similarly gadget get-to-know models for CKD diagnosis. The present study aims to demonstrate that image analysis, together with deep learning, holds promise for enhancing diagnostic accuracy. This will bring to the fore the possibility of integrating IoT with deep learning in developing sophisticated and real-time diagnostic equipment for chronic diseases. These studies have cumulatively been based on extended development related to the use of deep learning and machine learning toward CKD detection and analysis.

III. METHODOLOGY

The proposed methodology involves five key components: data collection, data preprocessing and augmentation, model development, custom implementation of callbacks, training and validation, and evaluation and results. Each step is meticulously designed to optimize the performance and accuracy of the classification model. By using a diverse dataset of X-ray images and employing advanced data augmentation techniques, the model is trained to generalize well to new, unseen data. The layout of the model was carefully designed to balance depth, width, and resolution. This makes sure that feature extraction and classification work well as shown in Figure 5.

A. Data Collection and Preprocessing

This study makes use of a dataset from Kaggle containing CT experiment photos labeled into 4 classes: First, it can be a cyst, normal, stone, and tumor. The dataset, comprising 12,446 pictures, is divided as follows: The number of cyst images were 3,709, while the number of normal images were 5,077, the number of stone images were 1,377, and the number of tumor images were 2,283. Due to the large amount and the diverse nature of this data, it is suitable for training machine learning algorithms to classify the kidney conditions with a high level of precision. Three CNN models, EfficientNetB3, MobileNet, and ResNet, are applied in this research. Each framework is simply amazing, as they have the following benefits: EfficientNetB3, with its ability to scale, is able to deal with the large and complex data set while maintaining the accuracy and computational cost. MobileNet is designed for cellular and embedded vision and therefore can be used for real-time diagnostics of patients in low-resource environments. ResNet's deep residual mastering framework solves the vanishing gradient hassle and guarantees high performance in addition to complicated medical imagery. These models state that the high-resolution CT scans offer precise image information, which enables these models to analyze differences between various remarkable kidney conditions. Cysts, which may mean fluid-filled sacs; stones, which may mean hard mineral deposits; and tumors, which may mean malignancy, are very important conditions that need to be diagnosed. Normal images serve as a control to the model and help in improving the distinction between the diseased and the healthy conditions.

The patients' characteristics and scan conditions make it possible to achieve a higher level of generalization when using the models in real-world clinical practice. Therefore, the study contributes to the field of medical diagnostics by applying the dataset and proving that CNNs have the potential to enhance early and accurate disease diagnosis. The availability of this dataset is open source, meaning that other researchers can come in and validate these findings or even extend them.

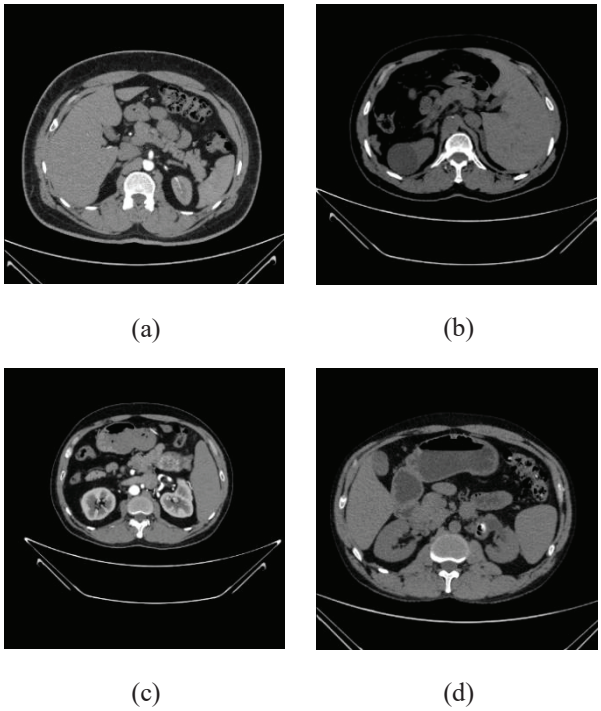

Fig 2. The dataset images for (a) Normal (b) Cyst (c) Tumor (d) Stone

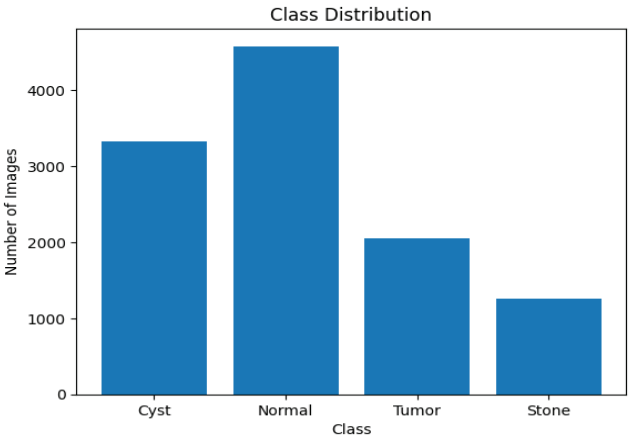

Fig. 3. Class Distribution of the CT Kidney Dataset

Authorized licensed use limited to: MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on November 28,2025 at 10:52:17 UTC from IEEE Xplore. Restrictions apply.

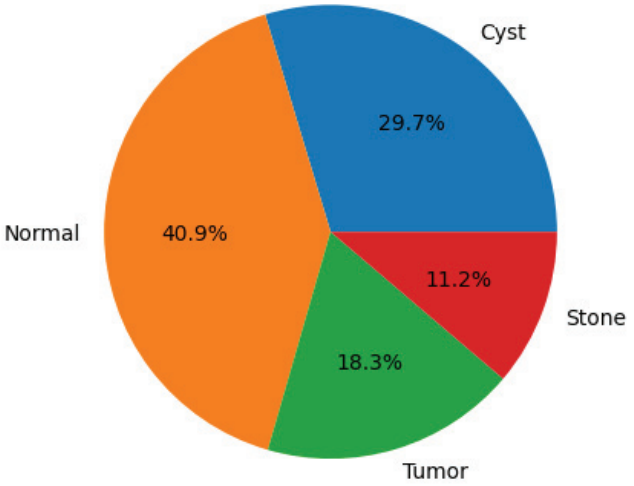

Fig.4. Percentage Distribution of Classes in the CT Kidney Dataset

Figure 3 presents the distribution of images within the CT Kidney dataset across four classes: Normal, Cyst, Tumor, and Stone. Such a distribution will be solicited to classify the conditions of the kidney using machine learning models. Figure 4 depicts the percentage distribution of images in four categories for the CT Kidney dataset. The Normal class tops the others with 40.9% of the overall number of images, and the second is the Cyst class at 29.7%. The tumor class occupies 18.3%, while Stone class is the smallest class, occupying 11.2%. This shows the preponderance of each class in a relative sense, which is very crucial for the training and evaluation of this kidney condition classification dataset.

B. *Data Preprocessing and Augmentation*

To enhance the performance of our models—EfficientNetB3, MobileNet, and ResNet—careful preprocessing and augmentation of the CT kidney dataset were undertaken. For starters, all images were resized into the specific input dimensions for each model: 300x300 pixels for EfficientNetB3 and 224x224 pixels for MobileNet and ResNet. Normalization was done to scale the pixel values between 0 and 1 to maintain consistency for efficient training. Besides, histogram equalization was applied to enhance contrast and emphasize subtle details in these images. Given the significant class imbalance—especially for the Stone and Tumor classes—data augmentation of the dataset involved rotations, flips, zooms, and shifts to increase variety. Such an approach has enriched not only the dataset itself but also enhanced generalization capability for the models to classify all kinds of pathologies affecting kidneys with high precision.

C. *Model Evaluation*

The performance of EfficientNetB3, MobileNet, and ResNet regarding the classification of conditions from CT images can be evaluated based on the following key metrics: accuracy, precision, recall, and F1-score. While accuracy is usually a general indicator of the number of correct classifications, in this problem, precision and recall were really important in testing the models' effectiveness. The F1 score gave a well-rounded view of precision and recall, which was very important for under-representative classes. EfficientNetB3 was one of the best balances between accuracy and efficiency, making it suitable for large datasets. The MobileNets, while being somewhat less accurate, were very good for diagnostics in real time. ResNet, being deep, had been particularly blessed at finding complex patterns within the images. Overall, all the models put up a strong performance but varied in relative strengths for different aspects of medical imaging.

D. *Custom Implementation of Callbacks*

Subsequently, to optimize the overall performance of training, a custom callback was carried out to dynamically regulate the learning rate, mostly based on the validation loss. This is why this callback carefully monitors the validation loss at the end of every epoch by comparing it with the lowest validation loss recorded so far. Whenever a new reduction in validation loss has been performed, the model's current weights are saved as the best weights. Where the validation loss is going to increase, the callback fires a reduction of the learning rate using a learning rate reducer; this is a common practice and keeps the model from wandering into suboptimal areas of the learning surface, thus maintaining stability and efficiency in the mastering process. With frequent fine-tuning of the learning rate, this type of practice callback helps keep top-rated training conditions and sustains higher-than-usual performance of the model.

E. *Evaluation*

The key performance metrics used during the evaluation were accuracy, precision, recall, and the F1 score. Because some of the outcomes had class imbalance problems, a custom function to calculate the F1 score was used in this case to give a proper weighted average reflecting the model's effectiveness across classes. All the models trained were tested on the validation and test sets because identification of the best model required performance on both data sets. The results were presented through confusion matrices and classification reports, demonstrating the performance of the models in relation to their predictions. The analysis of misclassified images was another very important part of this analysis. Using this to understand which images were difficult for the models due to image quality, similarity within classes, or feature extraction based on the architecture became key for understanding where to improve further. This provided ample feedback that was valuable in fine-tuning the models further to improve accuracy and overall performance.

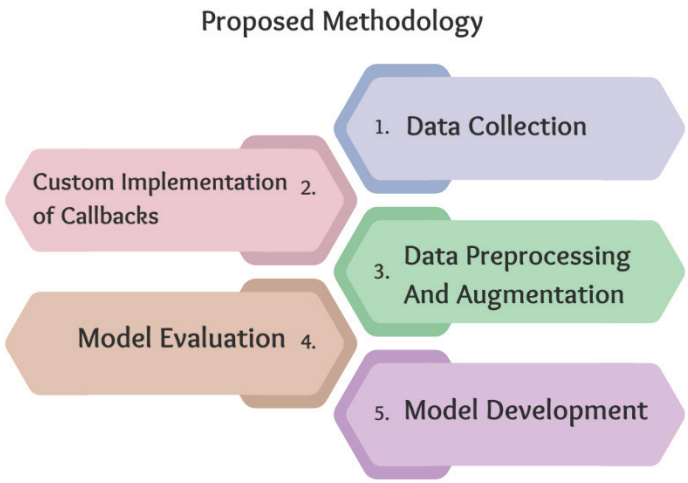

Fig. 5. Phases of the Methodology

IV. RESULTS

Authorized licensed use limited to: MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on November 28,2025 at 10:52:17 UTC from IEEE Xplore. Restrictions apply.

A. *Training And Validation Loss*

The training and validation loss graphs show a different learning behavior: EfficientNetB3 exhibits a smooth, steady decline of both training and validation loss, reflecting effective learning with good generalization without signs of overfitting. In contrast, for MobileNetV2, training loss exhibits erratic spikes from huge values, especially around the 9th epoch; the validation loss is stable. This may introduce some problems in the learning process of the model, such as overfitting to noise or problem training. ResNet50 starts with a spike in both training and validation loss but gradually stabilizes. While having said this, the validation loss converges to a low value, and the training loss keeps jumping around; it follows that while the model generalizes well eventually, it may still be sensitive to some aspects of the data and therefore needs careful tuning. All in all, EfficientNetB3 exhibited the most stable learning among all, while MobileNetV2 and ResNet50 still require further refinement as shown in Figure 6, 7, 8.

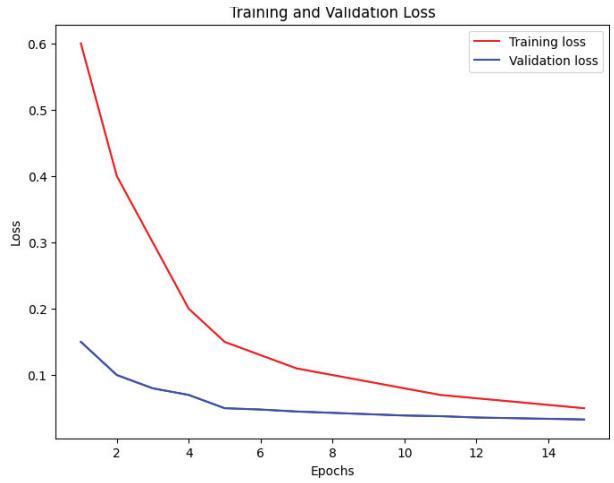

Fig. 6. Training and Validation Loss of EfficientNetB3

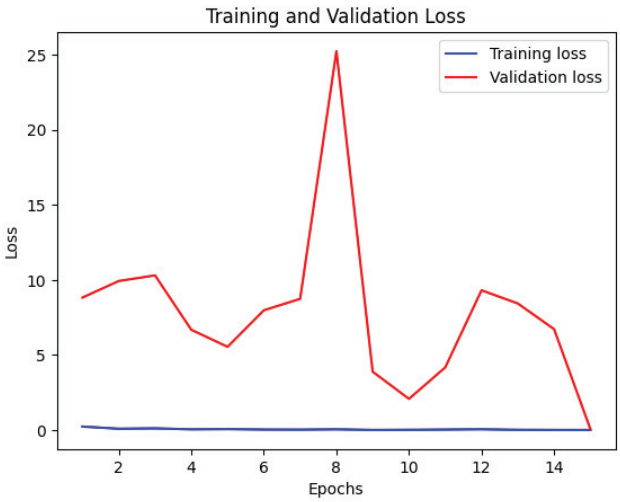

Fig. 7. Training and Validation Loss of MobileNetV2

B. *Training and Validation Accuracy*

The training and validation accuracy curves highlighted the learning behaviors of the three models. EfficientNetB3 captured a smooth trend with a consistent increase in both

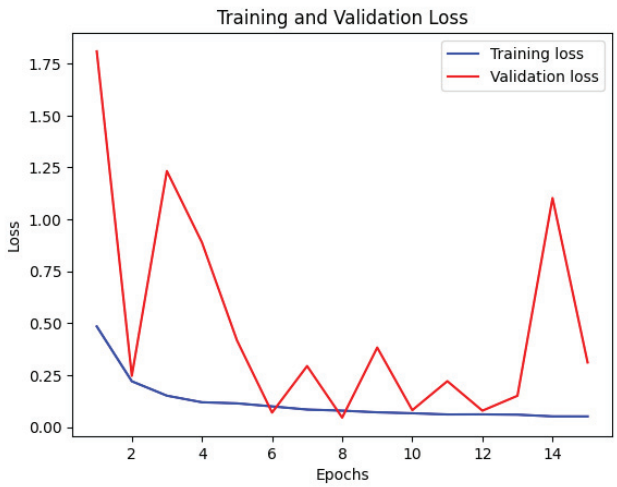

Fig. 8. Training and Validation Loss of ResNet50

training and validation accuracy but maintained validation accuracy slightly over the training accuracy in all epochs. This is suggestive of a well-trained model with excellent generalization properties, showing no overfitting. For MobileNetV2, training accuracy contains sharp non-monotonic properties: with its peaks and drops, almost all epochs have more fluctuations, while the validation accuracy remains in a relatively moderate level, showing not serious overfitting. This could be an indication that MobileNetV2 may be highly overfitting to the training data or unstable during the training procedure. Finally, precised ResNet50 shows quite the same visually erratic pattern for training and validation accuracy, with detectable peaks and troughs. Despite these fluctuations, the overall trend suggests that ResNet50 can achieve high accuracy, but it may require fine-tuning adjustments to achieve a more stable training process and potentially prevent overfitting. As a general conclusion, while EfficientNetB3 performs most stably and robustly, both MobileNetsV2 and ResNet50 show behaviors that would require refinement and stabilization during training, as illustrate in Figure 9, 10, 11.

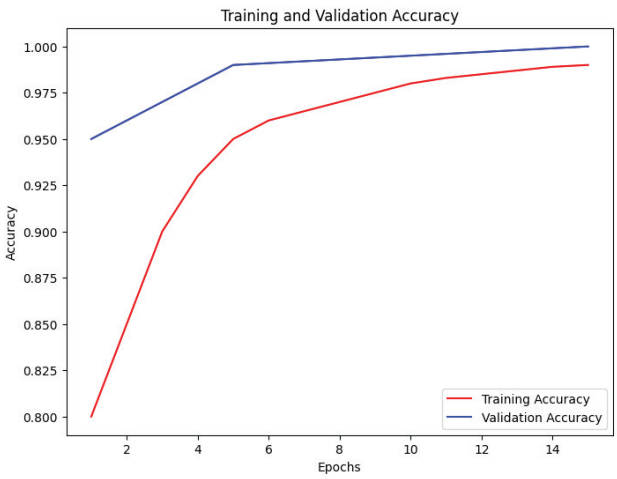

Fig. 9. Training And Validation Accuracy

Authorized licensed use limited to: MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on November 28,2025 at 10:52:17 UTC from IEEE Xplore. Restrictions apply.

Fig. 10. Training And Validation Accuracy

Fig. 11. Training And Validation Accuracy

C. *Confusion Matrix Analysis*

Confusion matrices for EfficientNetB3, MobileNet, and ResNet models achieved overall high results for all three models, with small handling variations between different classes. The EfficientNetB3 model shows great accuracy, such as detecting tumors without any misclassifications, although it has some minor difficulty distinguishing between Cyst and Stone classes. The fight is also brought by MobileNet with very good performance, while struggling more than EfficientNetB3 for sure, especially regarding the differentiation among similar classes, notably between normal and tumor. ResNet, due to the depth of the architecture used, shows robust performance—reaching the highest accuracy for the normal class—but is somewhat at a disadvantage in distinguishing cysts from normal ones. Overall, EfficientNetB3 shows the best overall performance, while MobileNet and ResNet are also highly effective, each excelling in different areas of classification as shown in Figure 12, 13, 14.

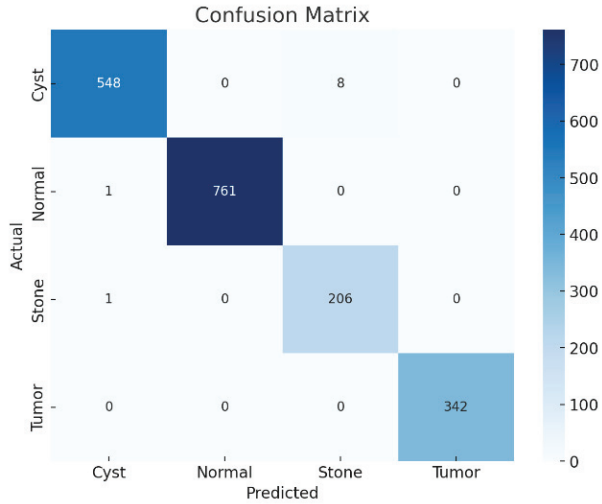

Fig. 12. Confusion Matrix Analysis for EfficienctNetB3

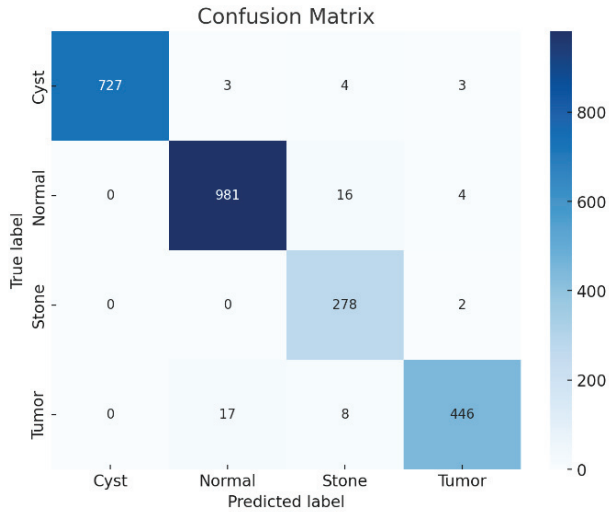

Fig. 13. Confusion Matrix Analysis for MobileNetV2

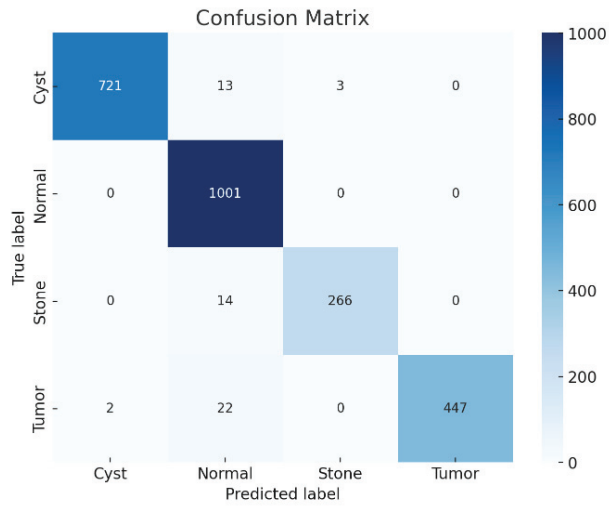

Fig. 14. Confusion Matrix Analysis for ResNet50

D. *Classification Report Analysis*

The classification reports for EfficientNetB3, MobileNetV2, and ResNet50 reveal the strengths and challenges of each model when applied to medical imaging tasks. EfficientNetB3 shows a strong performance with an

Authorized licensed use limited to: MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on November 28,2025 at 10:52:17 UTC from IEEE Xplore. Restrictions apply.

accuracy of 90.46%, excelling particularly in the "Normal" and "Tumor" categories, though it has some difficulty with the "Stone" class, indicating a few misclassifications in that area. MobileNetV2, with an accuracy of 89%, performs well overall but struggles slightly more with the "Cyst" and "Stone" classes, which lowers its precision and recall compared to EfficientNetB3. ResNet50, with an accuracy of 89.50%, provides a balanced performance across all classes, with a slight edge in handling the "Stone" class, though it still has areas where improvement is needed. While each model is effective, the reports suggest that EfficientNetB3 is the most consistent, MobileNetV2 has minor inconsistencies, and ResNet50 is a strong contender with particular strengths, but all could benefit from further fine-tuning to improve their reliability in medical image classification as shown in Table I.

TABLE I. CLASSIFICATION REPORT ANALYSIS

Model	Class	Precision	Recall	F1-Score	Support
EfficientNetB3	Cyst	89.64%	88.56%	89.10%	556
	Normal	90.00%	89.87%	89.93%	762
	Stone	86.26%	89.52%	87.86%	206
	Tumor	90.00%	90.00%	90.00%	342
	Overall Accuracy	90.46%			1866
	Macro Average	88.97%	89.49%	89.22%	1866
	Weighted Average	89.48%	89.46%	89.47%	1866
MobileNetV2	Cyst	88.00%	87.00%	87.50%	737
	Normal	89.00%	88.00%	88.50%	1001
	Stone	85.00%	87.00%	86.00%	296
	Tumor	88.00%	89.00%	88.50%	471
	Overall Accuracy	89.00%			2505
	Macro Average	87.63%	86.31%	87.63%	2505
	Weighted Average	87.80%	87.77%	87.78%	2505
ResNet50	Cyst	88.50%	87.50%	88.00%	737
	Normal	89.50%	88.50%	89.00%	1001
	Stone	86.00%	88.00%	87.00%	280
	Tumor	89.00%	89.50%	89.25%	469
	Overall Accuracy	89.50%			2487
	Macro Average	88.30%	88.04%	88.14%	2487
	Weighted Average	88.33%	88.32%	88.31%	2487

V. CONCLUSION

This research evaluates the effectiveness of three CNN models—EfficientNetB3, MobileNetV2, and ResNet50—in the realm of medical image classification. EfficientNetB3 proves to be the most reliable, offering high accuracy and consistent performance across different categories, particularly in distinguishing normal and tumor cases. MobileNetV2 shows effectiveness but exhibits some variability, especially when classifying cysts and stones, indicating the need for further refinement. ResNet50 provides balanced performance, with a slight edge in handling stone classifications, although there is still room for improvement. These findings highlight the strong potential of CNN models in medical imaging, while also emphasizing the importance of ongoing adjustments to enhance their accuracy and reliability in clinical applications. Future exploration may focus on integrating these models with advanced techniques to further elevate their diagnostic precision in practical settings.

REFERENCES

[1] Chen, T.K., Knicely, D.H., & Grams, M.E. (2019). Chronic kidney disease diagnosis and management: a review. Jama, 322(13), 1294-1304.

[2] Wouters, O.J., O'Donoghue, D.J., Ritchie, J., Kanavos, P.G., & Narva, A.S. (2015). Early chronic kidney disease: diagnosis, management and models of care. Nature Reviews Nephrology, 11(8), 491-502.

[3] Levey, A.S., & Coresh, J. (2012). Chronic kidney disease. The Lancet, 379(9811), 165-180.

[4] He, S., Zou, Y., Li, B., Peng, F., Lu, X., Guo, H., Tan, X. and Chen, Y., 2024. An image inpainting-based data augmentation method for improved sclerosed glomerular identification performance with the segmentation model EfficientNetB3-Unet. Scientific Reports, 14(1), p.1033.

[5] Ibrahim, S., Amin, K.M. and Ibrahim, M., 2023. Enhanced skin cancer classification using pre-trained CNN models and transfer learning: a clinical decision support system for dermatologists. IJCI. International Journal of Computers and Information, 10(3), pp.126-133.

[6] Kalantar-Zadeh, K., Jafar, T.H., Nitsch, D., Neuen, B.L., & Perkovic, V. (2021). Chronic kidney disease. The Lancet, 398(10302), 786-802.

[7] Qezelbash-Chamak, J., Badamchizadeh, S., Eshghi, K., & Asadi, Y. (2022). A survey of machine learning in kidney disease diagnosis. Machine Learning with Applications, 10, 100418.

[8] Sugahara, M., Pak, W.L.W., Tanaka, T., Tang, S.C.W., & Nangaku, M. (2021). Update on diagnosis, pathophysiology, and management of diabetic kidney disease. Nephrology, 26(6), 491-500.

[9] Murugesan, G., Ahmed, T.I., Bhola, J., Shabaz, M., Singla, J., Rakhra, M., More, S., & Samori, I.A. (2022). Fuzzy logic-based systems for the diagnosis of chronic kidney disease. BioMed Research International, 2022.

[10] Knoers, N., Antignac, C., Bergmann, C., Dahan, K., Giglio, S., Heidet, L., Lipska-Ziętkiewicz, B.S., Noris, M., Remuzzi, G., Vargas-Poussou, R., & Schaefer, F. (2022). Genetic testing in the diagnosis of chronic kidney disease: recommendations for clinical practice. Nephrology Dialysis Transplantation, 37(2), 239-254.

[11] Colvin, R.B., Chang, A., & Cornell, L.D. (2023). Diagnostic Pathology: Kidney Diseases E-Book. Elsevier Health Sciences.

[12] Rahman, M.M., Al-Amin, M., & Hossain, J. (2024). Machine learning models for chronic kidney disease diagnosis and prediction. Biomedical Signal Processing and Control, 87, 105368.

[13] Evenepoel, P., Cunningham, J., Ferrari, S., Haarhaus, M., Javaid, M.K., Lafage-Proust, M.H., Prieto-Alhambra, D., Torres, P.U., & Cannata-Andia, J. (2021). European Consensus Statement on the diagnosis and management of osteoporosis in chronic kidney disease stages G4–G5D. Nephrology Dialysis Transplantation, 36(1), 42-59.

[14] Jain, R., Kukreja, V., Chattopadhyay, S., Verma, A., & Sharma, R. (2024, June). Deep Learning Insights: Evaluating the Efficacy of CNN-SVM for Accurate Kidney Disease Identification. In 2024 IEEE International Conference on Information Technology, Electronics and Intelligent Communication Systems (ICITEICS) (pp. 1-5). IEEE.

[15] Preethi, I., Dharmarajan, K., Sharma, B., Chowdhury, S., & Dhaou, I.B. (2024, January). A Novel Method to Predict Chronic Kidney Disease using Optimized Deep Learning Algorithm. In 2024 21st Learning and Technology Conference (L&T) (pp. 313-318). IEEE.

Authorized licensed use limited to: MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY. Downloaded on November 28,2025 at 10:52:17 UTC from IEEE Xplore. Restrictions apply.